E-COMMERCE SALES & PROFITABILITY ANALYSIS

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1. Executive Summary

E-Commerce Sales & Profitability Analysis is a data analytics project designed to evaluate e-commerce sales performance, revenue, costs, profitability, product performance, customer activity, and regional sales.

The project follows an end-to-end data analytics workflow using Python for data analysis, MySQL for structured querying, and Power BI for interactive dashboard development.

The dataset contains 600 records and 13 columns. The objective is to transform transaction-level data into meaningful business insights that can help organizations understand sales trends, identify profitable products, compare regional performance, and support data-driven decisions.

Final business metrics and recommendations will be added after the dataset has been validated and the analysis completed.

2. Introduction

E-commerce businesses generate data through orders, customers, products, sales, and operational activities. Analyzing this data helps businesses understand customer purchasing behavior, evaluate revenue, and monitor profitability.

However, raw transaction data must be validated before it can be used for business reporting. This project applies data analytics techniques to prepare the dataset, calculate important KPIs, identify patterns, and present findings through a Power BI dashboard.

3. Problem Statement

E-commerce businesses need reliable information about sales revenue, costs, product performance, and profitability.

Without proper analysis, it can be difficult to identify top-performing products, compare regions, recognize changes in sales over time, and determine which areas require improvement.

This project aims to address these challenges by analyzing e-commerce transaction data and developing an interactive reporting solution.

4. Project Objectives

1. Understand the structure of the e-commerce dataset.
2. Validate data quality and prepare the data for analysis.
3. Calculate total sales, total cost, total profit, and profit margin.
4. Identify top-performing products and categories.
5. Compare sales performance across regions.
6. Analyze sales trends over time.
7. Use SQL to answer business-related questions.
8. Build an interactive Power BI dashboard.
9. Develop actionable business recommendations based on verified results.

5. Dataset Overview

Dataset file: `ecommerce_cleaned_data.csv.xlsx`

Total records: 600

Total columns: 13

The dataset will be used to analyze e-commerce transactions and business performance. The exact column names, data types, and definitions should be documented from the uploaded file before the final report is published.

The following information categories are relevant to the planned analysis, where available in the dataset:

- Order and transaction information
- Order dates
- Customer information
- Product and category information
- Regional information
- Sales and cost values
- Quantity and profitability measures

6. Tools and Technologies

Python: Used for loading, processing, validating, and analyzing data.

Pandas: Used for data manipulation, data quality checks, grouping, aggregation, and feature engineering.

Microsoft Excel: Used for spreadsheet inspection and validation.

MySQL: Used to store structured data and answer business questions through SQL queries.

Power BI: Used to create interactive KPI cards, charts, filters, and business dashboards.

GitHub: Used to maintain project files, document the workflow, and present the project in a professional portfolio.

7. Data Preparation and Validation

Before calculating business metrics, the dataset should be checked for:

- Missing values
- Duplicate records
- Incorrect data types
- Invalid or inconsistent dates
- Inconsistent product or category labels
- Missing or invalid sales and cost values
- Unusual numerical values
- Duplicate order identifiers where uniqueness is expected

The original dataset should be preserved. Any changes should be documented, and the validated dataset should be saved separately.

The dataset's filename includes both `.csv` and `.xlsx`. Its actual file format should be confirmed before choosing the appropriate loading method in Python.

8. Key Performance Indicators (KPIs)

The following KPIs will be calculated where the required fields are available.

Total Sales

The sum of valid sales values.

Total Cost

The sum of valid cost values.

Total Profit

Total Profit = Total Sales − Total Cost

Profit Margin (%)

Profit Margin (%) = (Total Profit ÷ Total Sales) × 100

Total Orders

The number of distinct valid order identifiers.

Total Quantity Sold

The sum of valid quantities sold.

Average Order Value

Average Order Value = Total Sales ÷ Number of Distinct Valid Orders

All KPI values will be calculated from validated data. Profit and margin calculations must use the actual definitions and fields available in the dataset.

9. Business Questions

The project aims to answer these questions:

1. What is the total sales revenue?
2. What are the total costs and total profit?
3. Which products generate the highest sales?
4. Which categories contribute the most revenue?
5. Which products have the highest profit margins?
6. Which regions perform best?
7. How do sales change over time?
8. What is the average order value?
9. Which products generate high sales but relatively low profit?
10. Which areas may offer opportunities to improve business performance?

The final answers will be supported by calculated results, tables, and charts.

10. Exploratory Data Analysis

Python and Pandas will be used to perform exploratory data analysis.

The planned analysis includes:

- Descriptive statistics

- Sales and cost summaries
- Product-wise performance
- Category-wise performance
- Region-wise sales comparisons
- Monthly sales trends
- Profitability analysis
- Investigation of unusual values and patterns

Appropriate visualizations will be selected after reviewing the available columns.

11. SQL Business Analysis

After data validation, the dataset will be imported into MySQL.

SQL queries will be developed to calculate and compare:

- Total sales, cost, and profit
- Product-wise sales performance
- Category-wise revenue
- Regional performance
- Monthly sales trends
- Order counts and average order value
- Profitability by product or category

SQL results will be compared with Python calculations to check consistency.

12. Power BI Dashboard

An interactive Power BI dashboard will be developed to present the main business metrics.

The planned dashboard may include:

- KPI cards for Total Sales, Total Cost, Total Profit, and Profit Margin
- Sales trend over time
- Product-wise sales and profit comparisons
- Category-wise performance
- Region-wise sales analysis
- Relevant slicers for date, product, category, and region

The final dashboard design will depend on the fields available in the dataset.

13. Project Workflow

1. Load and inspect the dataset.

2. Review data types and data quality.
3. Clean and validate the data.
4. Calculate profit and other relevant measures.
5. Perform exploratory data analysis in Python.
6. Load validated data into MySQL.
7. Write SQL queries to answer business questions.
8. Develop the Power BI dashboard.
9. Validate KPI calculations and summarize insights.
10. Publish the project documentation and permitted files on GitHub.

14. Expected Outcomes

The completed project is expected to deliver:

- A validated e-commerce dataset
- Reliable sales and profitability KPIs
- Product and category comparisons
- Regional sales analysis
- Sales trend visualizations
- SQL business queries
- An interactive Power BI dashboard
- Evidence-based business insights and recommendations

These are planned deliverables, not final findings.

15. Current Project Status

Dataset information provided:

- File: `ecommerce_cleaned_data.csv.xlsx`
- Rows: 600
- Columns: 13

Pending validation:

- Exact column names and data types
- Missing-value and duplicate checks
- Sales, cost, and profit calculations
- Product and regional analysis
- SQL queries and Power BI dashboard
- Final findings and business recommendations

16. Conclusion

E-Commerce Sales & Profitability Analysis demonstrates a structured approach to analyzing business transaction data.

By combining Python, SQL, and Power BI, the project aims to transform raw data into reliable KPIs, meaningful visualizations, and actionable business insights.

The final report will be updated after validating the dataset, completing the analysis, and confirming the results.
